

An introduction to Python language

Constructor

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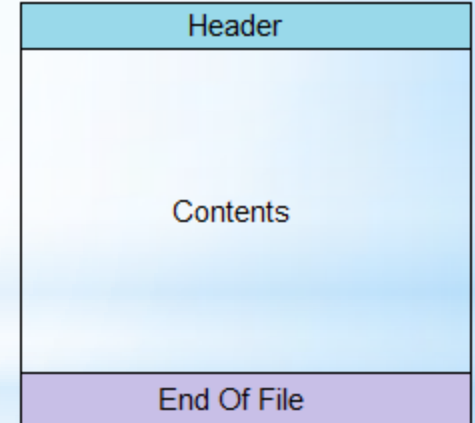
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Files (few terms)

file: is a contiguous set of bytes used to store data. This data is organized in a specific format and can be anything as simple as a text file or as complicated as a program executable.

Files on most modern file systems are composed of three main parts:

- ☐ **Header:** metadata about the contents of the file (file name, size, type,...)
- ☐ **Data:** contents of the file as written by the creator or editor
- ☐ **End of file (EOF):** special character that indicates the end of the file



Files (few terms)

Files in python:

❑ Text files

- Text file stores information in **ASCII OR UNICODE** character. In text file everything will be stored as a character.
- each line is terminated by special character called **EOL**. In text file some translation takes place when this EOL character is read or written. In python EOL is '\n' or '\r' or combination of both

❑ Binary files

- It stores the information in the same format as in the memory i.e. data is stored according to its data type so no translation occurs.
- In binary file there is no delimiter for a new line.
- Binary files are faster and easier for a program to read and write than text files.
- Data in binary files cannot be directly read, it can be read only through python program.

Files (few terms)

For accessing the file we need to three major parts:

Folder Path: folder location on the file system that contain the file

File Name: the actual name of the file

File Extension: the file type its it is few characters after final period.

Examples: Folders

F:\\Kufa University\\lectures Python\\lecture1.txt'

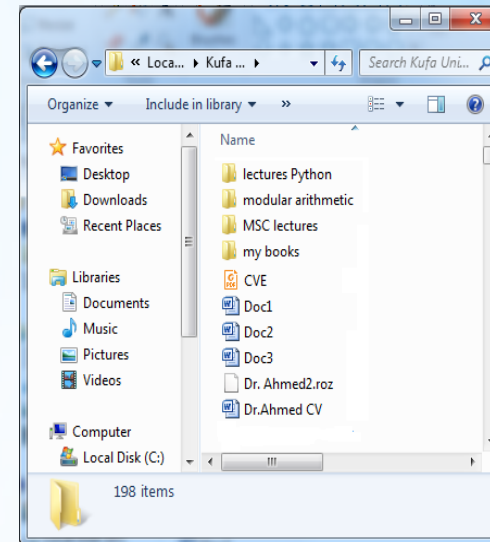
Folder path:

File Name: **File Extension:**

**Root directory
or
Drive**

Notes:

- Folder is container of files or other subfolders.
- A directory is a folder
- Some file extensions are: txt, ppt, pptx, doc, docx, pdf



Files (few terms)

Notes:

- Folder is container of files or other subfolders.
- A directory is a folder
- Some file extensions are: txt, ppt, pptx, doc, docx, pdf ...
- In python the full path can be written as:

'F:\\Kufa University\\lectures Python\\lecture1.txt'

'F:/Kufa University/lectures Python/lecture1.txt'

r'F:\Kufa University\lectures Python\lecture1.txt'

Files (its location)

For dealing with folders, we need to importing os module:

- ❑ Create folder:

`os.mkdir(path)`

- ❑ Delete empty folder:

`os.rmdir(path)`

- ❑ Getting all files and folders in a folder (level one):

`os.listdir(path)`

- ❑ Checking is file exist or not:

`os.path.isfile(path)`

- ❑ Checking is folder exist or not:

`os.path.isdir(path)`

Files (folders examples)

Ex1: Create folder with name “lecture” in the directory “C:\ ali” :



```
import os
os.mkdir("C:\\ali\\lecture")
```

Ex2: delete the folder “lecture” from the directory “C:\ ali” :



```
import os
os.rmdir("C:\\ali\\lecture")
```

Ex3: add folder “lecture” and inside it folder “1” in the directory “C:\ ali” :



```
import os
os.mkdir("C:\\ali\\lecture")
os.mkdir("C:\\ali\\lecture\\1")
```

Ex4: Get all files and folders in the directory “C:\ ali\lecture” in a list called L then separate the files in list F and folders in list D:



```
import os
mypath="C:\\ali\\lecture"
myfiles = os.listdir(mypath)
F=[]
D=[]
for f in myfiles:
    if os.path.isfile(mypath+"\\")+f):
        F.append(f)
    else:
        D.append(f)
```

Files (processing)

Steps in Data File Handling

☐ OPENING FILE:

We should first open the file for read or write by specifying the name of file and mode.

☐ PERFORMING READ/WRITE:

Once the file is opened now we can either read or write for which file is opened using various functions available

☐ CLOSING FILE

After performing operation we must close the file and release the file for other application to use it

Files (open and close file)

For open a file use open: `open(file, mode='r', buffering=-1, encoding=None, errors=None, newline=None, closefd=True, opener=None)`

The important arguments are:

- ❑ file: refer to the full path of file with its name
- ❑ Mode: the mode in which the file is opened (the default is 'r' read):
- ❑ Encoding : encoding is the name of the encoding used to decode or encode the file. This should only be used in text mode. For Arabic use “utf-8”, “utf-16”, or 'utf-8-sig'

For closing file use:

`Filename.close()`

It release the file and save the any write to physical disk.

Note: open function is built-in function used standalone while close() must be called through file handle